Chi-Sheng (Michael) Chen



Physiological Time-Series ML • Biosignal Foundation Models • LLM Agents • Clinical AI

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Research Affiliate at Harvard Medical School & BIDMC building interpretable ML for clinical biosignals (EEG, EMS trauma triage) and biosignal foundation models; EEG depression-treatment model in routine clinical use at Taipei VGH. Prior: CTO of Neuro Industry, Digital IC Engineer at MediaTek. Reviewer for NeurIPS/ICML/AAAI/KDD and 42 journals.

Education

National Taiwan University (NTU) — Taipei, Taiwan
M.S., Biomedical Electronics & Bioinformatics — Adv. Prof. Cheng-Ta Li, MD & Prof. Chung-Ping Chen — *Sep 2019 – Jun 2021*

National Chiao Tung University (NCTU) — Hsinchu, Taiwan
B.Eng. in Electrophysics & B.S. in Interdisciplinary Science (dual degree) — *Sep 2015 – Jun 2019*

Selected Experience

Harvard Medical School & Beth Israel Deaconess Medical Center — MA, USA
Research Affiliate (primary appointment) — Advisor: Prof. Gabriel Brat — *Nov 2024 – Now*
- Real-time EMS trauma-triage **multi-agent** pipeline: ASR (Whisper) → LLM-agent event extraction (RAG, tool use, task planning) → structured prediction over **4M+ patient records**.

WhaleForce (subsidiary of VICI Holdings) — Taipei, Taiwan
Fractional Token Efficiency Research Lead (Advisor) — *Aug 2026 – Now*
- Advising the AI × quant-trading division of Taiwan's leading HFT prop firm on LLM token-efficiency research, multi-agent architecture, and financial data extraction; **cut LLM token usage by 82%**.

Neuro Industry, Inc. — CA, USA
Co-Founder & CTO — *Mar 2024 – Jan 2025*
- Built GCP MLOps pipeline training on 10K+ EEG samples; led 3 papers on quantum ML and EEG-to-image diffusion.

Omnis Labs — Remote
Co-Founder — *Sep 2024 – May 2026*
- AI-driven DeFi liquidity/risk curator across AMM and perp DEX (Hyperliquid, Curve, Steer, Aster); 3 research papers.

MediaTek Inc. — Hsinchu, Taiwan
Digital IC R&D Engineer — *Dec 2021 – Oct 2023*
- Microprocessor IP for flagship 5G SoCs; hardware-virtualization RTL design & UVM/SystemVerilog verification.

Agent & LLM Engineering

Berkeley RDI AgentX-AgentBeats — 3rd Place / 1,300+ teams (2026). *MadGAA*, a multi-agent OSCE medical evaluation system: role-specialized agents (examiner, patient, judge) + verifiable scoring rubric (RAG + tool use + verifier).

US-stock LLM agent suite (deployed). Browser agent as an explicit PLAN→LOCATE→ACT→VERIFY state machine with fault-injection recovery; layered SEC 10-K extractor; 22 research agents + 11 placebo controls with lookahead-free backtests, per-provider cost ledger, and eval dashboards.

Robotics Workcell Optimizer (live). NL → spec → robot selection → 2D layout → 3D preview via multi-LLM abstraction (Claude/OpenAI/Gemini, validation-repair loop) + OR-Tools CP-SAT; ISO 13855 safety as inviolable hard constraints.

Also: *flyhypo* (connectome + PubMed + LLM → grounded, falsifiable neuron hypotheses); *paper-evidence* (verbatim-quote gate + cross-family judge to block LLM hallucination); two live NYCU RAG physics tutors (Next.js + Gemini + pgvector); OpenAI RLHF contract (SFT/DPO/reward modeling).

Stack: LangChain/LangGraph, ReAct & function calling, multi-agent orchestration, vLLM, LoRA/QLoRA, SFT/DPO/RLHF, RAG (pgvector/FAISS/Qdrant), OR-Tools, OpenAI/Anthropic/Gemini SDKs.

Selected Publications

(first/co-first author unless noted; 49 papers, 491 cites, h-index 13 — full list on Google Scholar)

Biosignals & Physiological Time-Series ML.

Chen, C.S., et al. "FreqLens: Interpretable Frequency Attribution for Time Series Forecasting." *Under review, ICLR 2027*; arXiv:2602.08768.

Chen, C.S., et al. "Large Cognition Model: Towards a Pretrained EEG Foundation Model." *arXiv:2502.17464* (2025).

Chen, C.S., et al. "A Unified SPD Token Transformer Framework for EEG Classification." *Under review, Neurocomputing*; arXiv:2601.21521.

Chen, C.S., Wei, C.S. "Mind's Eye: Image Recognition by EEG via Multimodal Similarity-Keeping Contrastive Learning." *arXiv:2406.16910* (2024) — 32 cites.

Chen, C.S., et al. "EEG Signal–Based Image Generation Using Diffusion Models." *JMIR Medical Informatics* (2025, IF 3.8).

Chen, C.S., Brat, G.A. "Measuring Browser Webcam Gaze Honestly: A Capture-Clock Methodology." *MICCAI DEMI Workshop* (2026, spotlight talk).

Li, C.T., **Chen, C.S.**, et al. "Prediction of Antidepressant Responses to Non-Invasive Brain Stimulation from Frontal EEG." *J. Affective Disorders* (2023, IF 6.5).

Quantum ML & Other.

Chen, C.S., et al. "Unraveling Quantum Environments: Transformer-Assisted Learning in Lindblad Dynamics." *Physical Review A* (2025).

Chen, C.S., et al. "QEEGNet: Quantum Machine Learning for Enhanced EEG Encoding." *J. Signal Processing Systems* (2025); SiPS 2024 — 65 cites.

Chen, C.S., Kuo, E.J. "Quantum Adaptive Self-Attention for Quantum Transformer Models." *Quantum Science and Technology* (2026).

Chen, C.S., et al. "Quantum Adaptive Self-Attention for Financial Rebalancing on AMMs in DeFi." *IEEE ICASSP* 2026 (oral).

Chen, C.S., et al. "Res-VMamba: Fine-Grained Food Classification via Selective State-Space Models with Deep Residual Learning." *arXiv:2402.15761* (2024); *PLoS ONE* (2025) — 83 cites.

Professional Service

Conference Reviewer: NeurIPS 2026, ICML 2026 (Gold Reviewer), AAAI 2027, KDD 2026, MICCAI 2026, ICASSP 2026, NLDL 2027, ACM ICAIF 2026.

Journal Reviewer (42 journals): incl. TPAMI, TNNLS, IEEE JBHI, National Science Review (OUP), npj Quantum Information, Scientific Reports, Elsevier Neural Networks. **PC:** QNRL @ IEEE WCCI 2026.